INCH-POUND
MIL-DTL-24643/73
9 October 2008

# DETAIL SPECIFICATION SHEET

## CABLE, ELECTRICAL, -20 °C TO + 90 °C, 1000 VOLTS, TYPE LSTCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

| | | |
|---|---|---|
| First | - | Copper conductor, tin-coated, size 200, 550 strands of 24AWG |
| Second | - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third | - | Thermoset insulation. Nominal wall thickness of 0.055 inch. Circuit identification shall be accomplished by solid-colored insulation or printed marking on the wire in accordance with table II. |
| Fourth | - | Three conductors cabled together with a lay in accordance with MIL-DTL-24643, paragraph 3.4.4. Fillers may be used as necessary to form a firm well-rounded assembly. |
| Fifth | - | A binder tape applied helically with overlap. |
| Sixth | - | Cross-linked polyolefin jacket. Color black. Nominal wall thickness of 0.075 inch. |



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TABLE I. Construction details.

| Military Part No. M24643/73 | Type & Size | AWG Conductor Size | Conductor Resistance (ohms) | Cable Jacket Thickness (min. avg.) (inch) | Minimum Bend Radius (inch) | Overall Diameter (inches) | |
|---|---|---|---|---|---|---|---|
| | | | | | | (min.) | (max.) |
| -01UN | LSTCF-200 | 0000 | 0.054 | 0.065 | 9 | 1.700 | 1.800 |

TABLE II. Circuit identification.

| Component | Identification |
|---|---|
| 1 | Black |
| 2 | White |
| 3 | Red |

REQUIREMENTS:

Qualification required.

EXAMINATION AND TESTS:

| | |
|---|---|
| Basic Electricals: | |
| Conductor resistance (ohms/1000 feet at 25 °C, max.) | See table I |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 3000 |
| Insulation resistance (megohms/1000 feet, min.) | |
| Conductor to conductor | 500 |
| Conductor continuity | No Failure |
| Group A: | |
| Visual and dimensional | No Failure |
| Group B: | |
| Cold bending cable | No Failure |
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 200 °C) | 50 |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength ($lb/in^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |
| Jacket (cable) | |



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| | |
|---|---|
| Tensile strength ($lb/in^2$, min.) | 1300 |
| Elongation (percent, min.) | 160 |
| Tear (pounds per inch thickness, min.) | 35 |
| Group C: | |
| Physicals (aged) | |
| Air oven | |
| Insulation (extruded) | |
| Tensile strength (percent of unaged, min.) | 75 |
| Elongation (percent of unaged, min.) | 75 |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (insulation) (method 1 only; cycles, min.) | 25 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Shrinkage | No Failure |
| Group D: | |
| Flame propagation (cable) | No Failure |
| Qualification Inspection: | |
| Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following: | |
| Aging and compatibility (cable) (125±5 °C) | No Failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 2 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Insulation | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |



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| | |
|---|---|
| Smoke index, max. | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index, max. | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket – Type A hardness, min.) | 80 |
| Electrical moisture absorption | No Failure |

UNIT ORDERING LENGTHS: 250 feet (nominal)

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-120)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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